SENTIENCE

Platform for Adaptive Mental Health Intervention

Capstone Project Proposal

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Mentor Consent Form

I hereby agree to be the mentor of the following Capstone Project Team

Project Title: Sentience		
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Project Overview

The proposed capstone project is designed as an AI-based digital platform where smart journaling is used to support long-term mental wellness and emotional self-reflection. In comparison to conventional text-based journaling systems, a more complete representation of emotional state is captured, since thoughts are not limited to text but can also be recorded through audio. Micro-expressions are detected using ResNet-50 to infer facial emotions, while vocal tone and fatigue are analyzed through models such as Wav2Vec 2.0. Privacy of personally identifiable information is maintained through on-device masking techniques implemented using MS Presidio, ensuring that sensitive data is not exposed. Based on the identified emotional state, customised reflection prompts are generated following cognitive behavioural therapy (CBT) principles, so that deeper and more structured self-reflection can be encouraged.

Beyond simple journaling, the platform is positioned as a searchable personal memory system where past emotions and experiences can be revisited and understood over time. Natural language queries can be used to retrieve relevant past entries, making reflection more intuitive and less effort-driven. Patterns in emotional behaviour are gradually identified by linking feelings with habits, events, and lifestyle factors. At the same time, a comprehensive health dashboard is provided where wearable data such as sleep patterns and heart rate are combined with mood trends, giving a more complete picture of well-being. A detailed "emotion scoreboard" is presented to visualise emotional variations, and predictive models like LSTM and ARIMA are applied to detect possible signs of emotional instability, though predictions may not always be perfectly accurate in every case.

To ensure continued engagement, consistent interaction is encouraged through personalised nudges, simple gamification elements, and context-aware conversational support. Behavioural patterns are learned over time so that timely check-ins can be triggered during commonly observed emotional low points. Ambiguous feelings are handled through an intent recognition layer, where clarifying questions are generated and responses are translated into more structured emotional states. With user consent, a dedicated clinician interface is made available where summarised trends and alerts for significant behavioural changes can be accessed. Overall, the system is intended to function as a supportive tool that helps individuals understand their emotions better, while still recognising that professional mental health care plays an essential and irreplaceable role.

Need Analysis

In today's fast-paced environment, ongoing stress, anxiety, and emotional fatigue are being experienced by a growing number of individuals, largely driven by academic pressure, work demands, and constant digital exposure. Despite how common this has become, meaningful emotional reflection is often not practiced, usually due to lack of time, low motivation, or simply absence of a safe space for expression. Traditional journaling tools are available, but they are often dependent on plain text and consistent effort, which many people struggle to maintain over time. Because of this, important emotional signals tend to be overlooked until they gradually turn into more serious concerns. This situation makes it clear that a system is needed where emotional experiences can be captured and understood in a more passive and natural way.

If a typical week is considered, especially for someone balancing studies and work, it can be observed that stress slowly builds up, sleep quality gets affected, and small frustrations begin to grow into anxiety. These changes are rarely noticed in real time, as gradual emotional shifts are often smoothed out by memory. However, if such experiences are consistently recorded and analyzed in an intelligent manner, early patterns could be identified much sooner. By combining multimodal journaling with machine learning-based analysis, a clearer understanding of emotional behaviour can be developed. Over time, links between daily habits, physical health, and emotional states can be uncovered, giving individuals a more structured way to understand themselves, even if not every insight is always perfectly accurate.

This system is therefore intended to go beyond being just a diary. It is designed as something that quietly observes how emotions evolve over time and helps in noticing when stress or unhealthy patterns begin to emerge. Timely nudges and small suggestions are provided when needed, so that emotional responses can be managed a little better in everyday life. In situations where immediate professional help is not easily accessible, such a system can offer early support and awareness. It is not meant to replace therapy, but rather to act as a first step—something that helps individuals understand themselves better and, when needed, move towards professional care with more clarity.

Literature Survey

Digital Phenotyping & Multimodal Sensing

As shown in prior systematic reviews, mental health signals obtained from smartphones and wearable devices have been validated for tracking behavioural and psychological patterns [1]. The concept of *digital phenotyping* has been introduced to describe how passively collected data—such as sleep, activity, heart rate variability, and voice—can be combined to monitor mental health conditions over time. It has also been demonstrated that multimodal emotion recognition, where multiple data sources are combined, achieves higher accuracy compared to single-modality approaches [5]. For example, speech prosody alone has been shown to account for a significant portion of depression-related variance, but detection performance is further improved when facial and physiological signals are integrated alongside it [2], [3].

NLP and Affective Models

Language analysis has been widely established as a key component in understanding emotional and mental states. Prior work has shown that deep learning models trained on behavioural and linguistic data can achieve high accuracy in detecting mental health conditions [4]. These approaches are leveraged in the system, where journal entries are processed using advanced NLP models to extract emotional and cognitive markers. Based on this analysis, context-aware insights are generated, enabling a more structured interpretation of user experiences.

Virtual Agents and Empathy

[4] Affective computing literature highlights the importance of adapting to the user's emotion in a virtual therapist. Emotion-aware agents can dynamically adapt to a user's affective state, which can lead to better engagement and outcome. [6] Non-verbal cues are also very important in a virtual therapist. For example, incorporating eye contact, facial mimicry, and nodding in a virtual therapist increased users' feelings of understanding and trust. [7] Empirical studies have also demonstrated that avatars providing rich verbal and non-verbal cues can lead to higher social presence and empathy. Emotional contagion is also well-documented in the literature, whereby users tend to mirror their avatar's emotions. For example, avatars mimicking users' emotions can lead to users mimicking their avatar's emotions, thus enhancing the sense of rapport. However, animations must be well-synchronized with speech, or they will be seen as unnatural. Empirical studies have demonstrated that avatars that are more human-like and attractive are rated higher on trust. In conclusion, a lifelike embodied agent that can respond to a user's affective cues can lead to a better therapy experience.

Privacy-Preserving AI

As mental health data is extremely sensitive in nature, the system ensures on-device privacy. This is achieved using techniques such as data minimization and anonymization, ensuring only the required amount of data is processed while maintaining the standard approach towards ensuring privacy.

MetaHuman Avatars

[6] For the operationalization of an empathic agent, a high-fidelity avatar framework is used. The aim of the avatar is to facilitate a sense of embodied presence, and slight facial expressions and gestures, such as eye gaze, nodding, and body postures, help in creating a sense of empathy and engagement. Research has shown that when a counselor avatar nods or smiles at a critical moment, users feel understood. The use of such avatars is a great way of providing a strong nonverbal cue for the AI therapist.

Objectives

- A multimodal journaling system, which can effectively detect emotional inputs using text, voice will be developed. In addition, the emotional signals will be analyzed using machine learning models to better understand the emotions of the user.
- A semantic memory retrieval system will be developed using the Retrieval-Augmented Generation (RAG) framework. This will allow the user's previous journal entries to be effectively searched through and significant emotional trends and patterns, which emerge in the user's journaling sessions over a period of time, to be identified.
- A well-being dashboard, which effectively integrates emotional information and other critical physiological parameters like sleep and heart rate, will be developed. This will allow the long-term relationships between the user's lifestyle and emotional well-being to be easily visualized.
- Supportive interventions, which are proactive and timely, will be facilitated by the development of predictive analytics models. These models will effectively identify the early signs of emotional instability or stress trends in the user.

Methodology

Stage 1: Data Capture and Privacy Handling

An end-to-end emotional intelligence system is designed to support multimodal journaling through a privacy-first approach. Users can record entries in text, voice, after which sensitive information is immediately anonymized using a lightweight on-device PII detection pipeline. Voice inputs are transcribed using optimized Whisper models. Text inputs are analyzed using fine-tuned transformer models to extract emotional and cognitive signals. Instead of relying on generic benchmarks, a personalized baseline is continuously learned from each user’s historical data to better reflect individual emotional patterns.

Stage 2: Multimodal Analysis and Interpretation

Each input modality generates an emotion probability vector, which is combined using a late fusion mechanism with interpretable weights. This allows transparency in understanding how different signals contribute to the final emotional state. To handle vague expressions, a transformer-based intent layer is used to generate clarifying questions and refine emotional inputs. Based on this understanding, CBT-aligned prompts are produced to guide self-reflection without prescribing actions. Additional features such as the “Unsent Letter” space allow users to process unresolved emotions, with lightweight analysis identifying recurring themes. Wearable data is integrated to align emotional trends with physiological patterns, while predictive models monitor sustained changes to flag potential emotional decline.

Stage 3: Memory, Insights, and Safety Mechanisms

All processed sessions are stored as embeddings in a secure vector database, enabling natural language queries over past emotional states and helping users identify long-term patterns. These insights are presented as suggestions rather than fixed conclusions. Safety mechanisms are incorporated to detect prolonged distress, triggering a “Red Zone” response that surfaces support resources or alerts trusted contacts with prior consent. A clinician-facing dashboard is also provided, where anonymized trends and alerts can be reviewed securely.

Stage 4: Engagement and System Design

Through adaptive reminders and an easy-to-use resilience tracking system, users are encouraged to engage consistently. An interface that communicates insights naturally, but lightweight so that it does not impose overhead on the system. AiVLC works towards achieving the end goal of overall architecture that is modular in nature, allowing scope for consistent improvement based on the feedback of the users and

evaluation metrics. It also has a clear positioning statement which identifies it as a tool that supports decision making, and does not intend to replace professional care.

Project Outcomes & Individual Roles

Expected Project Outcomes

- A cross-platform multimodal journaling application is to be developed, with support provided for text, voice, and wearable data inputs alongside an end-to-end pipeline for secure ingestion, on-device preprocessing, and emotion inference
- Fine-tuned machine learning models for text, audio, and physiological data are to be implemented, while a modular fusion framework is established to ensure the generation of a unified and interpretable emotional state
- A data and analytics layer is to be delivered, featuring a vector database with retrieval-augmented generation (RAG) capabilities, a well-being dashboard for trend visualization, and predictive models for the early detection of stress patterns.
- Comprehensive technical documentation and evaluation reports are to be produced, with performance metrics, system architecture, and privacy-preserving design covered to demonstrate the validity and real-world applicability of the system.

Individual Team Member Roles

Table 1: Individual Team Member Roles and Responsibilities

Name	Responsibilities	Module
Naman Jain	Develops core intelligence using MentalBERT, Wav2Vec, and LSTM models, with PII masking and LLM integration for multimodal emotion insights.	AI, NLP & Signal Processing
Gurnoor Singh	Develop React Native mobile UI/UX; integrate Google Fit/HealthKit; handle user interaction and journaling interface; display feedback dashboards	Frontend & Product Lead
Tanish Singla	Set up server and PostgreSQL database; implement RAG pipelines; develop API endpoints for data flow; ensure data security in transit	Backend & Data Engineer
Laksha Singla	Develops core intelligence by fine-tuning MentalBERT; implements PII masking; processes speech with Wav2Vec; builds LSTM models; integrates ML pipelines and GenAI LLMs for multimodal analysis	AI, NLP & Signal Processing
Hemant Jindal	Develop facial expression recognition using ResNet-50; implement late-fusion strategy combining text and audio; handle MetaHuman integration	Vision & Multimodal Fusion Lead

Work Plan

Table 2: Project Work Plan and Development Timeline

Phase	Activity	Timeline	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
1	System Architecture & Text Model Integration	Plan/ Actual										
2	Audio Pipeline (Whisper) & Wearable API Auth	Plan/ Actual										
3	Vector Database Setup & RAG Memory Bank	Plan/ Actual										
4	Multimodal Fusion & Predictive Forecasting (ARIMA/LSTM)	Plan/ Actual										
5	Dual-UI Development (Patient App & Clinician Portal)	Plan/ Actual										
6	System Integration, Testing & Latency	Plan/ Actual										
7	Final Polish, Demo Prep & Video Sandbox	Plan/ Actual										

Course Subjects

The following course subjects form the conceptual and practical foundation for the successful execution of this capstone project:

Natural Language Processing (NLP) – Advanced text classification for the 28-emotion detection model, semantic embedding generation, Audio-to-Text transcription pipelines (Whisper), and Named Entity Recognition (NER) for PII scrubbing.

Machine Learning – Time-series forecasting algorithms (such as ARIMA or LSTMs) applied to historical sentiment data to predict future emotional trends and mood fluctuations.

Deep Learning and Neural Networks – Transformer-based architectures (e.g., MentalBERT, LLMs) for deep contextual sentiment analysis and the implementation of Retrieval-Augmented Generation (RAG) pipelines.

Database Management Systems – Implementation of Vector Databases (e.g., Pinecone/ChromaDB) for the "Memory Bank" semantic search, alongside traditional databases for secure user routing and Role-Based Access Control (RBAC).

Mobile & Wearable Computing (or IoT) – API integration, authentication, and data normalization for physiological signals acquired via third-party wearables (Apple Health / Google Fit).

Human-Computer Interaction (HCI) – Asymmetric UI/UX design principles for developing a dual-sided platform (Patient Journaling App vs. Data-heavy Psychologist Dashboard), including clinical guardrails and "Red Zone" alerts.

Information Security and Privacy – Applied data masking techniques (Microsoft Presidio) to ensure secure, anonymized data flow between the user input, the AI processing layer, and the clinician portal

Software Engineering – End-to-end system design, modular backend architecture (FastAPI), RESTful API integration, version control, and agile methodology for integrating multi-modal data streams.

References

[1] Elgendi, M., Markov, K., Liu, H., De Vos, M., Khalaf, K., Khandoker, A., ... & Menon, C. (2026). Wearable devices for anxiety assessment: a systematic review. Communications Medicine, 6(1), 20.

[2] Jiang, Z., Seyedi, S., Griner, E., Abbasi, A., Bahrami Rad, A., Kwon, H., ... & Clifford, G. D. (2023). Multimodal mental health assessment with remote interviews using facial, vocal, linguistic, and cardiovascular patterns. medRxiv, 2023-09.

[3] Yang, Y., Fairbairn, C., & Cohn, J. F. (2012). Detecting depression severity from vocal prosody. IEEE transactions on affective computing, 4(2), 142-150.

[4] Schlicher, M., Li, Y., Murthy, S. M. K., Sun, Q., & Schuller, B. W. (2025). Emotionally adaptive support: a narrative review of affective computing for mental health. Frontiers in Digital Health, 7, 1657031.

[5] Pepino, L., Riera, P., & Ferrer, L. (2021). Emotion recognition from speech using wav2vec 2.0 embeddings. arXiv preprint arXiv:2104.03502.

[6] Choi, D. S., Park, J., Loeser, M., & Seo, K. (2024). Improving counseling effectiveness with virtual counselors through nonverbal compassion involving eye contact, facial mimicry, and head-nodding. Scientific Reports, 14(1), 506.

[7] Paiva, A., Leite, I., Boukricha, H., & Wachsmuth, I. (2017). Empathy in virtual agents and robots: A survey. ACM Transactions on Interactive Intelligent Systems (TiiS), 7(3), 1-40.